

Septicemia due to *Bacillus clausii* after the use of probiotics. A complication to keep in mind.

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Probiotics are live microorganisms that benefit the host in different clinical situations. *Bacillus clausii* is one of the most frequently used, but it is not without risk. To date, there are few reports of complications secondary to this agent in pediatric patients. To describe the case of an infant who developed after treatment sepsis due to *Bacillus clausii*. A 4-month-old female infant of indigenous ethnicity, from a rural area in the interior of Panama, 3 hours away from the nearest health sub-center by canoe, and with protein-calorie malnutrition, presented with acute diarrhea and moderate-severe dehydration, receiving Enterogermina as part of the initial treatment. She was transferred to a tertiary hospital, where she arrived with impaired consciousness, respiratory distress, and signs of shock. The initial blood culture reported growth of methicillin-resistant *Staphylococcus aureus* (MRSA), the gastrointestinal panel was positive for *Clostridioides difficile*, and later serial blood cultures of peripheral blood and central venous catheter confirmed growth of *Bacillus clausii*. With a torpid evolution and resistance to multiple antibiotic schemes, she died due to multisystem organ failure twelve days after admission. The use of probiotics as concomitant treatment in patients with some degree of immunosuppression should be administered with caution, considering the presence of risk criteria for complications such as malnutrition or intestinal epithelial damage due to severe diarrhea since they predispose to the development of bacteremia and/or sepsis.

Content Analysis of Commercially Available Probiotics.

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We carried out content analysis of four batches each of 3 commercially available probiotic formulations of *Bacillus clausii*. Species identification was done using MALDI-TOF-MS technique while bacterial count was done using plate colony count. Only one of the three probiotic formulation analyzed was found to have homogeneous population of *B. clausii* while none was found to have the exact viable bacterial count as suggested on the label.

Characterization of lipoteichoic acid structures from three probiotic *Bacillus* strains: involvement of D-alanine in their biological activity.

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Probiotics represent a potential strategy to influence the host's immune system thereby modulating immune response. Lipoteichoic Acid (LTA) is a major immune-stimulating component of Gram-positive cell envelopes. This amphiphilic polymer, anchored in the cytoplasmic membrane by means of its glycolipid component, typically consists of a poly (glycerol-phosphate) chain with D-alanine and/or glycosyl substitutions. LTA is known to stimulate macrophages in vitro, leading to secretion of inflammatory mediators such as Nitric Oxide (NO). This study investigates the structure-activity relationship of purified LTA from three probiotic *Bacillus* strains (*Bacillus cereus* CH, *Bacillus subtilis* CU1 and *Bacillus clausii* O/C). LTAs were extracted from bacterial cultures and purified. Chemical modification by means of hydrolysis at pH 8.5 was performed to remove D-alanine. The molecular structure of native

and modified LTAs was determined by ^1H NMR and GC-MS, and their inflammatory potential investigated by measuring NO production by RAW 264.7 macrophages. Structural analysis revealed several differences between the newly characterized LTAs, mainly relating to their D-alanylation rates and poly (glycerol-phosphate) chain length. We observed induction of NO production by LTAs from *B. subtilis* and *B. clausii*, whereas weaker NO production was observed with *B. cereus*. LTA dealanylation abrogated NO production independently of the glycolipid component, suggesting that immunomodulatory potential depends on D-alanine substitutions. D-alanine may control the spatial configuration of LTAs and their recognition by cell receptors. Knowledge of molecular mechanisms behind the immunomodulatory abilities of probiotics is essential to optimize their use.

Therapeutic Prospective of a Spore-Forming Probiotic-Bacillus clausii UBBC07 Against Acetaminophen-Induced Uremia in Rats.

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To screen *Bacillus clausii* UBBC07 as a putative probiotic strain and to examine the protective effect of probiotic-B. *clausii* UBBC07 spore on uremia on rats induced by acetaminophen. In vitro tests performed to screen potential probiotic strains were gastric and bile acid resistance and ability to reduce pathogen adhesion to surfaces. An in vivo study was performed on rats ($n = 18$) which were randomly divided into three groups: group I, control-receives normal food and water, groups II and III receive acetaminophen i.p. at the dose of 550 mg/kg/day for 10 days, groups III was treated with *B. clausii* UBBC07 at a dose of 1×10^9 CFU/day for 15 days. Urea, creatinine, malondialdehyde (MDA), and GSH levels and antioxidant enzymes like super oxide dismutase (SOD) and catalase activity were considered to analyze renal failure. Plasma urea and creatinine levels ($p < 0.05$) significantly increase and SOD, catalase, and GSH activity level significantly decrease in group II as compared with the control group. After treatment with probiotic, there was a significant increase in SOD and catalase ($p < 0.05$) and a significant decrease in serum urea, creatinine, and MDA ($p < 0.05$) in group III in response to group II. The results also revealed that probiotic was able to tolerate pH 3.0-9.0 and 0.3% bile salt. The present study suggests that *B. clausii* UBBC07 could be used as a novel alternative natural therapy for uremia, a major syndrome of CKD.

Investigation of Enterogermina's Protective and Restorative Mechanisms on the Gut Microbiota with PPI, Using SHIME Technology.

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Proton pump inhibitors (PPIs) are commonly prescribed medications associated with changes in the gut microbiome and dysbiosis when used long-term. Probiotics, such as Enterogermina® (containing four strains of *Bacillus clausii*) reduce side effects from triple therapy with PPI+antibiotics. We aim to assess the ability of this probiotic in preventing and/or treating the dysbiosis induced by PPI use. Faecal samples from six healthy donors were used to colonise a Triple-Mucosal-Simulator of the Human Intestinal Microbial Ecosystem® model with added ileal compartment. Changes in the microbial community composition and metabolite production were measured for PPI alone (control), PPI+Enterogermina (preventative), and Enterogermina treatment after PPI (curative). Differences were assessed by one-way ANOVA with Tukey's multiple comparisons test. The model was shown to replicate some of the effects of long-term PPI use. There were significant changes in microbial diversity and an increase in butyrate

levels in the preventative and curative arms, indicative of a beneficial effect to gut health. Probiotic use countered some of the effects of PPI use: *Streptococcus bovis* levels increased in the control arm but reduced following probiotic treatment. These results show that probiotic treatment with *B. clausii* may have beneficial effects on the gut microbiota following PPI treatment.

Alkalihalobacillus clausii (formerly Bacillus clausii) spores lessen antibiotic-induced intestinal injury and reshape gut microbiota composition in mice.

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The antibiotic-induced intestinal injury (AIJ) is associated with diarrhoea and gastrointestinal discomfort. However, the pathological intestinal mechanisms and related side effects associated with antibiotic use/misuse may be counteracted by probiotics. This study aims to evaluate the effect and the protective mechanisms of a probiotic formulation containing *Alkalihalobacillus clausii* (formerly *Bacillus clausii*; BC) spores in an experimental model of AIJ. C57/Bl6J mice were orally challenged with a high dose of ceftriaxone for five days along with BC treatment which lasted up to the 15th day. Our results showed the beneficial effect of the probiotic in preserving colonic integrity and limiting tissue inflammation and immune cell infiltration in AIJ mice. BC increased tight junction expression and regulated the unbalanced production of colonic pro- and anti-inflammatory cytokines, converging toward the full resolution of the intestinal damage. These findings were supported by the histological evaluation of the intestinal mucosa, suggesting a potential restoration of mucus production. Notably, BC treatment increased gene transcription of the secretory products responsible for epithelium repair and mucus synthesis and normalized the expression of antimicrobial peptides involved in immune activation. Reconstruction of complex and diverse gut microbiota in antibiotic-induced dysbiosis was recorded upon BC supplementation. Specifically, the expansion of *A. clausii*, *Prevotella rara* and *Eubacterium ruminantium* drove intestinal microbiota rebalance by primarily impacting Bacteroidota members. Taken together, our data indicate that BC administration alleviates AIJ by multiple converging mechanisms leading to restoring gut integrity and homeostasis and reshaping microbiota composition.

Kinetics of Intestinal Presence of Spores Following Oral Administration of Bacillus clausii Formulations: Three Single-Centre, Crossover, Randomised, Open-Label Studies.

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BACKGROUND AND OBJECTIVE: Probiotics are live microorganisms that may provide benefits including the prevention of gastrointestinal disorders and other diseases. Enterogermina is a probiotic mix of spores from four strains of *Bacillus clausii* (O/C, T, N/R and SIN), available in several oral formulations. The objective of this analysis was to evaluate and compare the kinetic profiles of different formulations of Enterogermina-vial [E4 once daily (OD) and E2 twice daily (BID)], capsule [EC2 three times daily (TID)], oral powder for suspension (ES6 OD) and oral powder not requiring suspension (E6 OD) from two studies from 2012 (EUDRACT 2010-024497-19 and 2010-023187-41) and one study from 2016 (EUDRACT 2015-003330-27). *B. clausii* spores were counted in homogenised faecal samples (results expressed as counts per gram) or after culture at 37 °C for 24-36 h (results expressed as colony-forming units). Kinetics were assessed by area under the concentration-time curve (AUC), maximum concentration (Cmax), time to maximum concentration (Tmax) and spore presence/persistence. In total, 22 subjects in

each of the 2012 studies and 30 subjects in the 2016 study were randomised (mean age 25.0-33.8 years across studies). The mean ($\pm$ SD) absolute faecal spore counts (in millions) expressed as AUC per hour were 270.7 ± 147.7 (E2 BID) and 213.8 ± 60.2 (E4 OD) in 2012 EGKINETIC4, 312.7 ± 218.0 (EC2 TID) and 319.0 ± 221.1 (ES6 OD) in 2012 EGKINETIC6, and 212.6 ± 118.0 (E6 OD) and 293.2 ± 247.2 (ES6 OD) in 2016 EGKINETIC6OP. The kinetic profiles of the different formulations of Enterogermina were similar, with superimposable AUC and daily curve profiles in each study up to the 8th day post dose. *B. clausii* spore presence/persistence in the intestine of healthy volunteers did not differ between the two formulations within each of the three studies. Enterogermina was well tolerated across all formulations and studies. These results show different formulations of Enterogermina had similar kinetic profiles within each study; however, they also showed that probiotics could be associated with high variability. The European Medicines Agency guidelines are the current bioequivalence reference, although only the Tmax parameter is used for high variability drugs. Due to the specific kinetics of probiotics, new parameters of bioequivalence could be necessary, considering, for example, variability via a parameter such as AUC. EUDRACT 2010-024497-19, 2010-023187-41 and 2015-003330-27.

Efficacy and Safety of the Adjuvant Use of Probiotic *Bacillus clausii* Strains in Pediatric Irritable Bowel Syndrome: A Randomized, Double-Blind, Placebo-Controlled Study.

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Current irritable bowel syndrome (IBS) treatments have limited efficacy and probiotics like *Bacillus clausii* (*B. clausii*) were found to be effective in the management of several gastrointestinal disorders. This phase III trial assessed the efficacy and safety of adding *B. clausii* (four strains: O/C, N/R, SIN, T), versus placebo, to conventional treatment of pediatric IBS in Mexico. Patients aged 6-17 years 11 months with IBS (Rome IV) for at least 2 months were randomized to receive either *B. clausii* (oral suspension, total dose 4 billion spores/day) or placebo once daily for 8 weeks. All patients also received conventional treatment. The primary endpoint was the difference in the proportion of patients with clinical improvements at Week 8 (Global Assessment Questions [GAQ]). Secondary endpoints included responders by Subject's Global Assessment of Relief for Children with IBS (SGARC); number/consistency of stools; abdominal distention/bloating; abdominal pain/intensity; and IBS behavior. 73.6% (95% confidence interval [CI] 67.3-80.0; *B. clausii* n = 129) and 78.5% (95% CI 72.5-84.4; placebo n = 130) of patients had symptom improvement ($p = 0.8182$). For Week 8 SGARC, 19.2% (*B. clausii*) and 20.9% (placebo) reported complete symptom relief. Stool evaluations, bloating, abdominal pain/intensity, and IBS behavior were similar between groups. Both treatments were well tolerated. No significant differences in efficacy between *B. clausii* and placebo were demonstrated in addition to conventional treatment. The sample size calculation was based on an expected placebo/conventional treatment response of 30-40%. However, the actual treatment response observed was 80% and, thus, a study with larger population would be warranted. In addition, this study was conducted during the COVID-19 pandemic, when such controlled social conditions may have resulted in better diet, greater family stability, less psychological stress, and lower risk of infections exacerbating IBS, thereby improving symptoms in both groups. 2018-004519-31.

Optimization of the treatment of rotavirus infection in children by using *bacillus clausii*.

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Introduction: Rotavirus infection is a leading place in the structure of acute intestinal infections in children. Rotavirus is excreted in 40-60 % of children hospitalized with gastroenteritis all over the world. Every year, 2 million patients are hospitalized with a severe form of RVI, 25 million need medical help from a doctor and 111 million cases are treated at home. **The aim:** The purpose of our study was to optimize the treatment of rotavirus infection in children by using *Bacillus clausii*. **Materials and methods:** There were 65 children with a rotavirus infection under supervision. The control group was consisted of 28 practically healthy children. The study of humoral immunity was carried out on the basis of determining the serum content of immunoglobulin G, immunoglobulin M, immunoglobulin A, and secretory immunoglobulin A in coprofiltrate. In the process of treatment, the children were divided into two groups: the first received standard treatment, the second group were added to standard treatment with a probiotic drug (*Bacillus clausii*). **Results:** In children with RVI with modified treatment main symptoms were reduced compared with the children receiving standard treatment, ($p < 0.001$). In patients with rotavirus infection in the acute period of the disease, a decrease in the concentration of IgA ($p < 0.001$) and an increase in IgM ($p < 0.001$) in serum and a decrease in sIgA ($p < 0.001$) in coprofiltrate was observed in comparison with children in control group. In the period of convalescence in children after the traditional treatment, it wasn't revealed normalization of the immunoglobulins. Patients receiving a probiotic drug in addition to traditional treatment it was revealed normalization of the parameters of serum immunoglobulins A, M, G and sIgA in coprofiltrate. **Conclusions:** So, the probiotic drug containing *Bacillus clausii* has a positive effect on the humoral immune system in children with rotavirus infection.

Resistance of mosquitoes to Lambda-Cyhalothrin and DDT in a Niger Delta Region of Nigeria.

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The escalation of insecticide resistance across the World Health Organization (WHO) African region calls for the routine monitoring of insecticide resistance in mosquitoes. While pyrethroids are recommended for use and dichlorodiphenyltrichloroethane (DDT) has been prescribed, it is imperative to reevaluate their efficacy across diverse geographical settings. The extent of resistance among mosquitoes to DDT and lambda-cyhalothrin remains understudied in Ika North East, Delta State, Nigeria, where the yellow fever virus was reported in 2019. We investigated the susceptibility of female mosquitoes to DDT (4%) and lambda-cyhalothrin (0.05%) in this specific study area. Adult female mosquitoes of *Anopheles gambiae*, *Culex quinquefasciatus*, and *Aedes albopictus*, aged between 2 and 4 days, were subjected to these insecticides utilizing the WHO bioassay method. The assessment of knockdown was done between 10 and 60 min, then mortality after 24 h. *C. quinquefasciatus* and *A. albopictus* mosquitoes exposed to DDT and lambda-cyhalothrin exhibited high susceptibility, resulting in complete mortality (100%); however, *A. gambiae* displayed resistance, with mortality rates of 19% and 76%, respectively. Notably, *A. gambiae* mosquitoes exposed to piperonyl butoxide (PBO)-DDT and PBO-lambda-cyhalothrin exhibited enhanced mortality, reaching 95% (indicating suspected resistance) and 100% (indicating susceptibility), respectively. Knockdown time (KDT) for 50% in mosquitoes exposed to lambda-cyhalothrin ranged from 3.94 to 33.51 min. Similarly, KDT model for 95% ranged from 19.04 to 84.15 min. Among the tested mosquito species, *Culex* mosquitoes exhibited the shortest knockdown resistance time for lambda-cyhalothrin, recorded at 3.94 min. Similarly, the KDT for DDT ranged from 24.97 to 187.06 min for 50% mortality and from 61.04 to 431.03 min for 95% mortality, respectively. *Anopheles* mosquitoes exposed to PBO + DDT recorded the lowest KDT. Our study underscores the potential of exercising caution in the use of DDT and lambda-cyhalothrin insecticides for the control of mosquitoes due to

emerging resistance.
